



DATA-537 PROJECT REPORT

By: Grant Gonnerman



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Introduction

Millions of fans from around the world tune into NFL football games every week. There seems to be no limit to the growth the NFL has seen in recent years, expanding into other countries for international games, and setting new viewership records during holiday games. AWS along with many other companies have increased their focus into the NFL providing real time statistics and player information adding another layer to the game. In this project I will look at weekly team statistics with the goal of identifying the most important features to predict team wins and losses.

Data Description

I collected this data myself from Yahoo sports who shows weekly player statistics. There are no ethical concerns with this data because it is all publicly available with no confidential information. I saved this data into excel files and aggregated them into team totals with python. This dataset was collected from weeks 1-13 of the 2025 NFL season and contains defense, kicking, kickoffs, passing, punting, receiving, returns, rushing, and win/loss data. There are 416 rows and 79 columns in this dataset and 191 missing values coming from columns where a statistic simply did not happen such as no field goals being taken between 0-19 yards by a team.

Column Name	Description
pass_completions	Number of completed passes by the quarterback
pass_attempts	Number of pass attempts thrown
passing_yards	Total yards gained through passing
passing_touchdowns	Touchdowns scored via passing plays
interceptions_thrown	Number of passes intercepted by the defense
passing_first_downs	Number of first downs gained by passing
times_sacked	Number of times the quarterback was sacked

sack_yards_lost	Total yards lost due to sacks
pass_fumbles	Number of fumbles on passing plays
pass_fumbles_lost	Fumbles on passing plays that were lost to defense
completion_percentage	Percentage of pass attempts completed
yards_per_attempt	Average yards gained per attempted pass
rush_attempts	Number of rushing attempts
rushing_yards	Total yards gained by rushing
rushing_touchdowns	Touchdowns scored through rushing plays
rushing_first_downs	First downs gained by rushing
rush_fumbles	Number of fumbles on rushing plays
rush_fumbles_lost	Fumbles on rushing plays that were lost
longest_rush	Longest rushing play recorded
yards_per_rush	Average yards gained per rush attempt
receptions	Number of passes caught
targets	Number of pass attempts thrown toward the receiver
receiving_yards	Total yards gained through receiving
receiving_touchdowns	Touchdowns scored on receptions
receiving_first_downs	First downs gained via receptions
receiving_fumbles	Fumbles on completed receptions
receiving_fumbles_lost	Receiving fumbles that were lost
yards_per_reception	Average yards gained per catch
fg_made_(0-19)	Field goals made from 0–19 yards
fg_attempted_(0-19)	Field goal attempts from 0–19 yards
fg_made_(20-29)	Field goals made from 20–29 yards
fg_attempted_(20-29)	Field goal attempts from 20–29 yards
fg_made_(30-39)	Field goals made from 30–39 yards
fg_attempted_(30-39)	Field goal attempts from 30–39 yards
fg_made_(40-49)	Field goals made from 40–49 yards

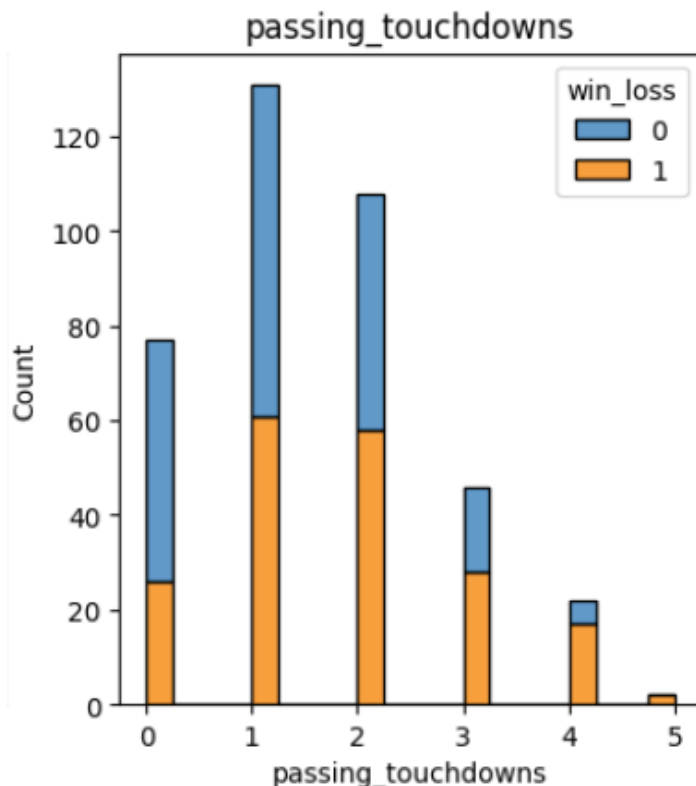
fg_attempted_(40-49)	Field goal attempts from 40–49 yards
fg_made_(50+)	Field goals made from 50 yards or more
fg_attempted_(50+)	Field goal attempts from 50 yards or more
fg_made_(total)	Total field goals made
fg_attempted_(total)	Total field goal attempts
extra_points_made	Extra points made after touchdowns
extra_points_attempted	Extra point attempts after touchdowns
kickoffs	Number of kickoffs made
kickoff_yards	Total yards on all kickoffs
kickoff_fair_catches	Fair catches made on kickoffs
touchbacks	Number of touchbacks on kickoffs
kick_returns_allowed	Number of kickoff returns allowed by coverage team
kick_return_touchdowns_allowed	Touchdowns allowed on kickoff returns
kick_return_yards_allowed_per_return	Average yards allowed per kickoff return
punts	Number of punts
punt_yards	Total yards gained on punts
punts_inside_20	Number of punts that landed inside the opponent's 20-yard line
punt_fair_catches	Number of fair catches on punts
punt_touchbacks	Touchbacks resulting from punts
blocked_punts	Number of punts blocked by defense
punt_returns_allowed	Punt returns allowed by coverage team
punt_return_yards_allowed_per_return	Average yards allowed per punt return
kick_returns	Number of kick returns by the team
kick_return_yards	Total yards gained on kick returns

kick_return_touchdowns	Touchdowns scored on kick returns
punt_returns	Number of punt returns
average_punt_return_yards	Average yards gained per punt return
solo_tackles	Number of solo tackles
assisted_tackles	Number of assisted tackles
total_tackles	Total tackles (solo + assisted)
sacks	Number of quarterback sacks
sack_yards	Yards lost by opponent due to sacks
run_stuffs	Number of run plays stopped at or behind line of scrimmage
interceptions	Passes intercepted by defense
interception_return_yards	Yards gained on interception returns
interception_touchdowns	Touchdowns scored on interception returns
forced_fumbles	Fumbles forced on opponents
fumble_return_touchdowns	Touchdowns scored on fumble recoveries
passes_defended	Pass breakups or deflections
safeties	Safeties scored by the defense
team_name_full	Full team name (e.g., “Arizona Cardinals”)
team_abbr	Team abbreviation (e.g., “ARI”)
win_loss	Indicates whether the team won or lost the game (0=loss, 1=win)
game_week	Week number of the NFL season

Exploratory Data Analysis

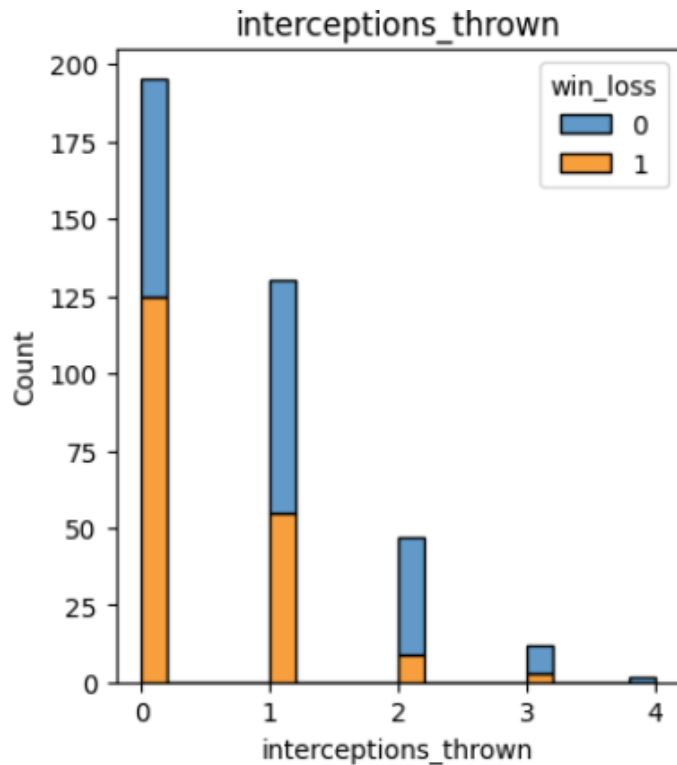
To get a better understanding of the data, I created a series of visualizations to explore the distribution of team statistics. The goal of this analysis is to identify key statistics that could be the most important factors in predicting wins and losses in the NFL.

Figure 1: Distribution of Passing Touchdowns



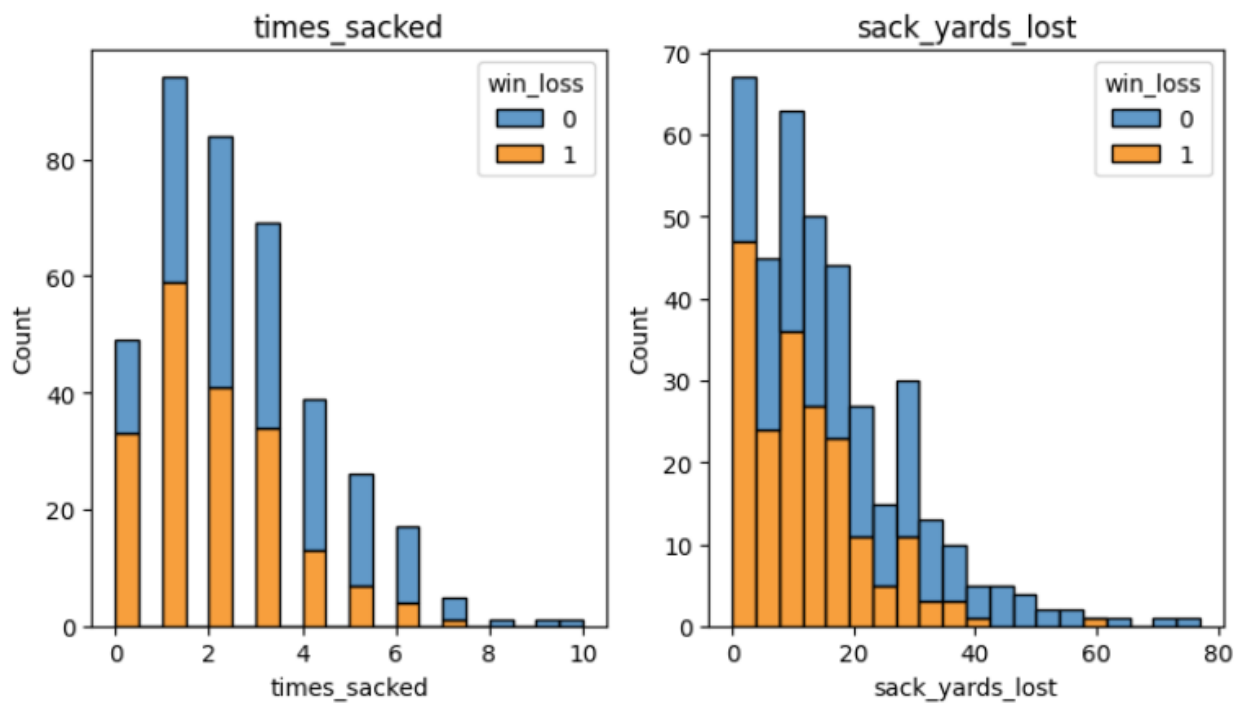
In the Figure to the left we can see the distribution of passing touchdowns. The first thing I noticed was the proportion of win to loss favored the loss class for lower values and the win class for higher values of passing touchdowns.

Figure 2: Distribution of Interceptions Thrown



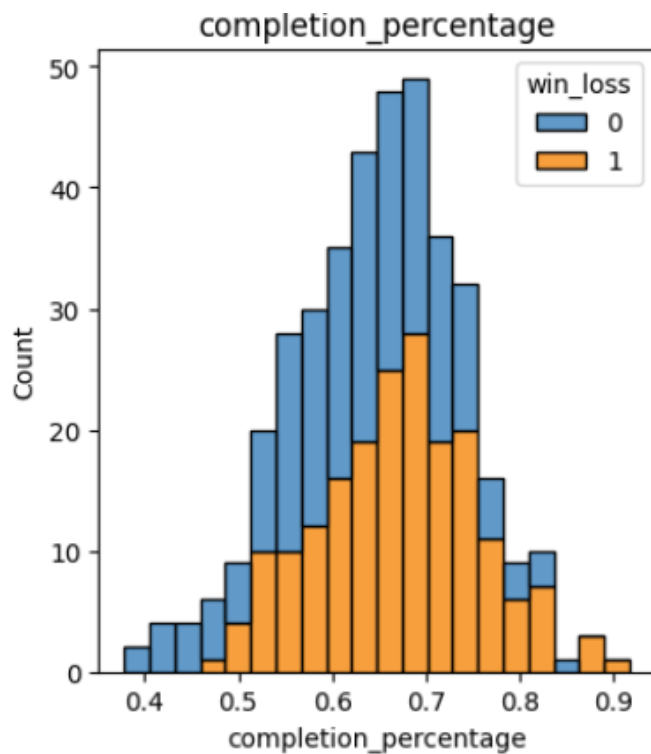
In the figure to the left of interceptions thrown by the quarterback we can see that as the number of interceptions thrown increases the proportion of the loss class increases.

Figure 3: Distribution of Sack Statistics



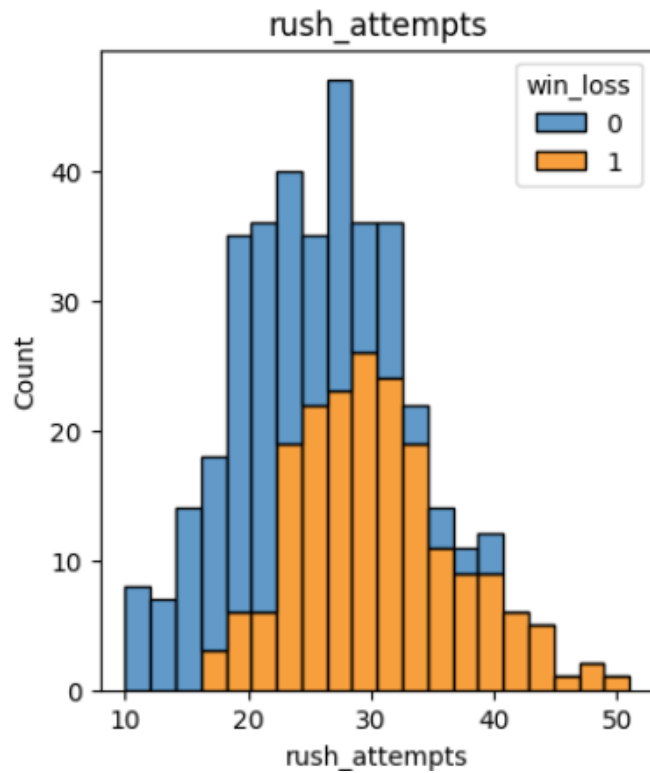
In the figure above we can see the distribution of the number of times the quarterback was sacked and how many yards were lost on those sacks. The main thing I noticed was that the loss class maximum values are higher than the win class and that the tail for the loss class is much longer than the win class.

Figure 4: Distribution of Completion Percentage



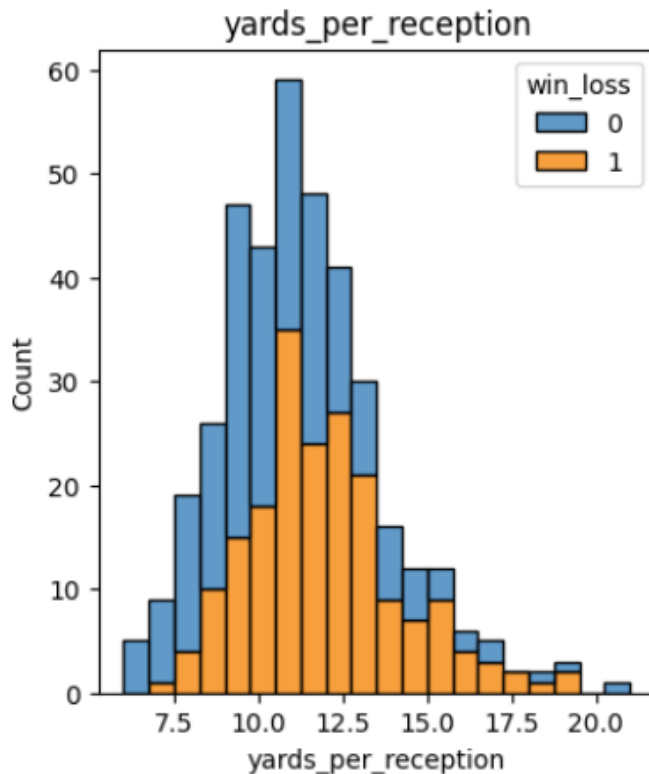
The figure to the left shows the completion percentage. Notice there are very few win occurrences below .5 and very few loss occurrences above .8.

Figure 5: Distribution of Rush Attempts



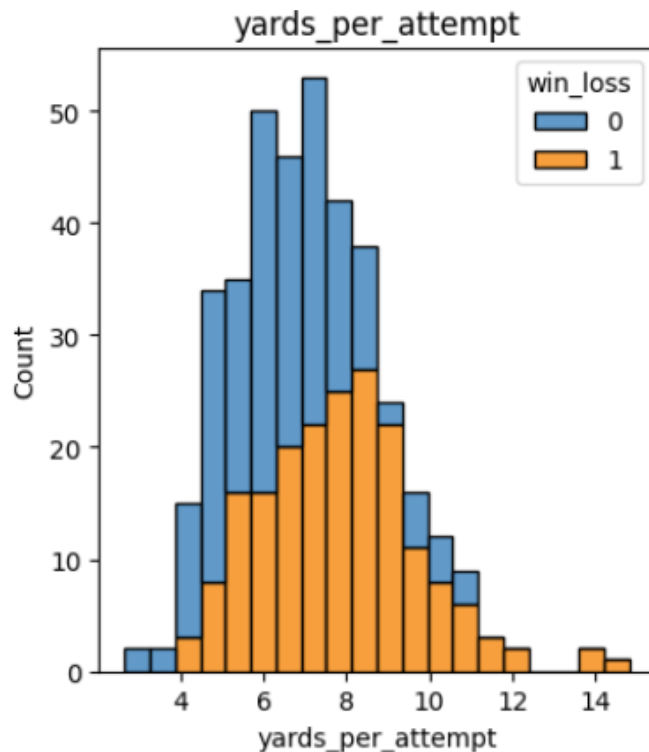
The figure to the left shows rush attempts per game. The shape is similar to the completion percentages chart showing the win class is distributed around 30 attempts whereas the loss class is distributed around 25 attempts with very few over 32.

Figure 6: Distribution of Yards per Reception



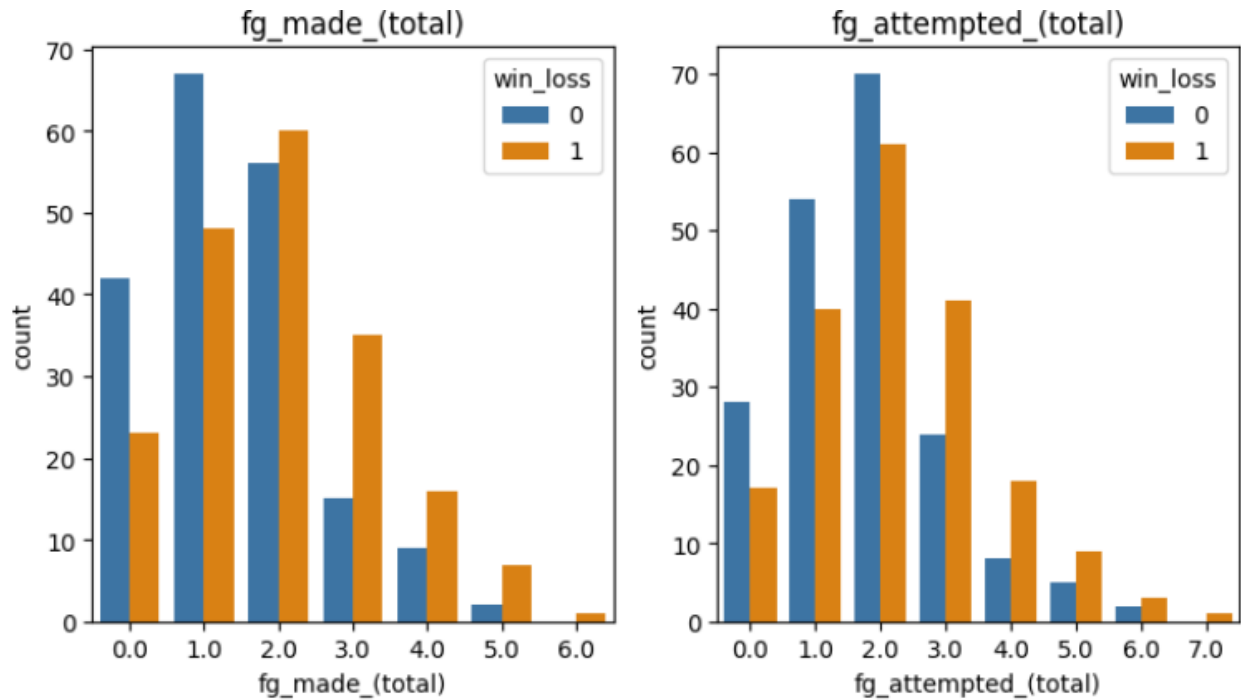
The chart to the left shows yards per reception. Both classes show similar distributions except for on the tails. The lower yards per reception show more losses while the higher yards per reception have more wins.

Figure 7: Distribution of Yards per Attempt



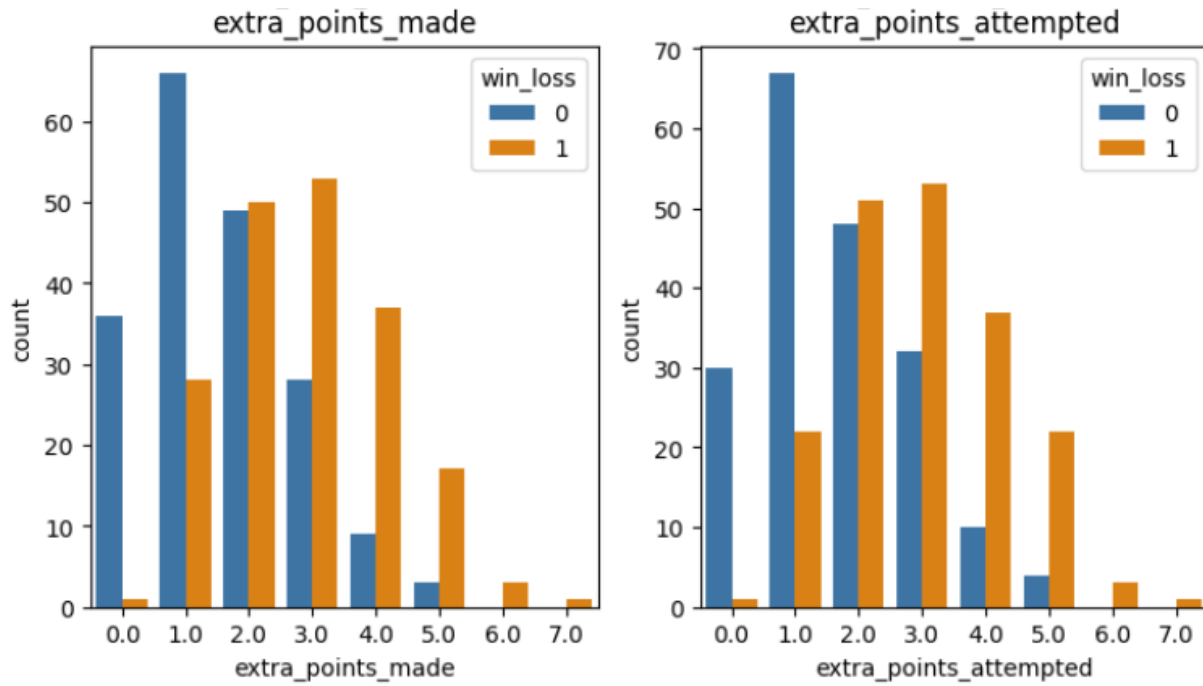
The chart to the left shows yards per attempt. The loss class is slightly skewed right distributed around 7 yards per attempt while the win class is more normally distributed around 8.5 yards per attempt.

Figure 8: Distributions of Field Goals Made and Attempted



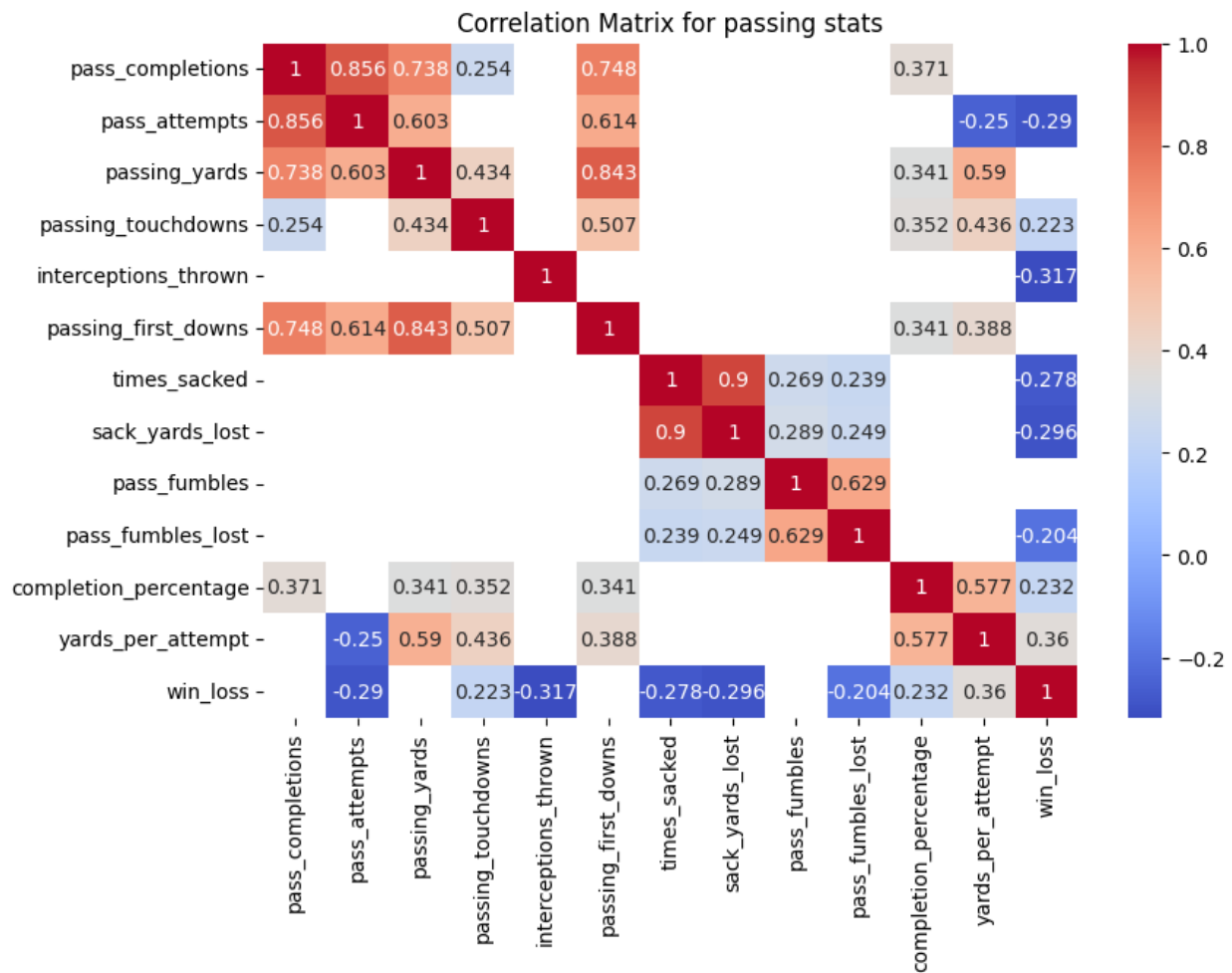
The charts above show field goals made and field goals attempted. The main thing I noticed was is the loss class is more prevalent for one field goals made and lower and two field goals attempted and lower. Anything above that the win classes are more prevalent.

Figure 9: Distribution of Extra Points Made and Attempted



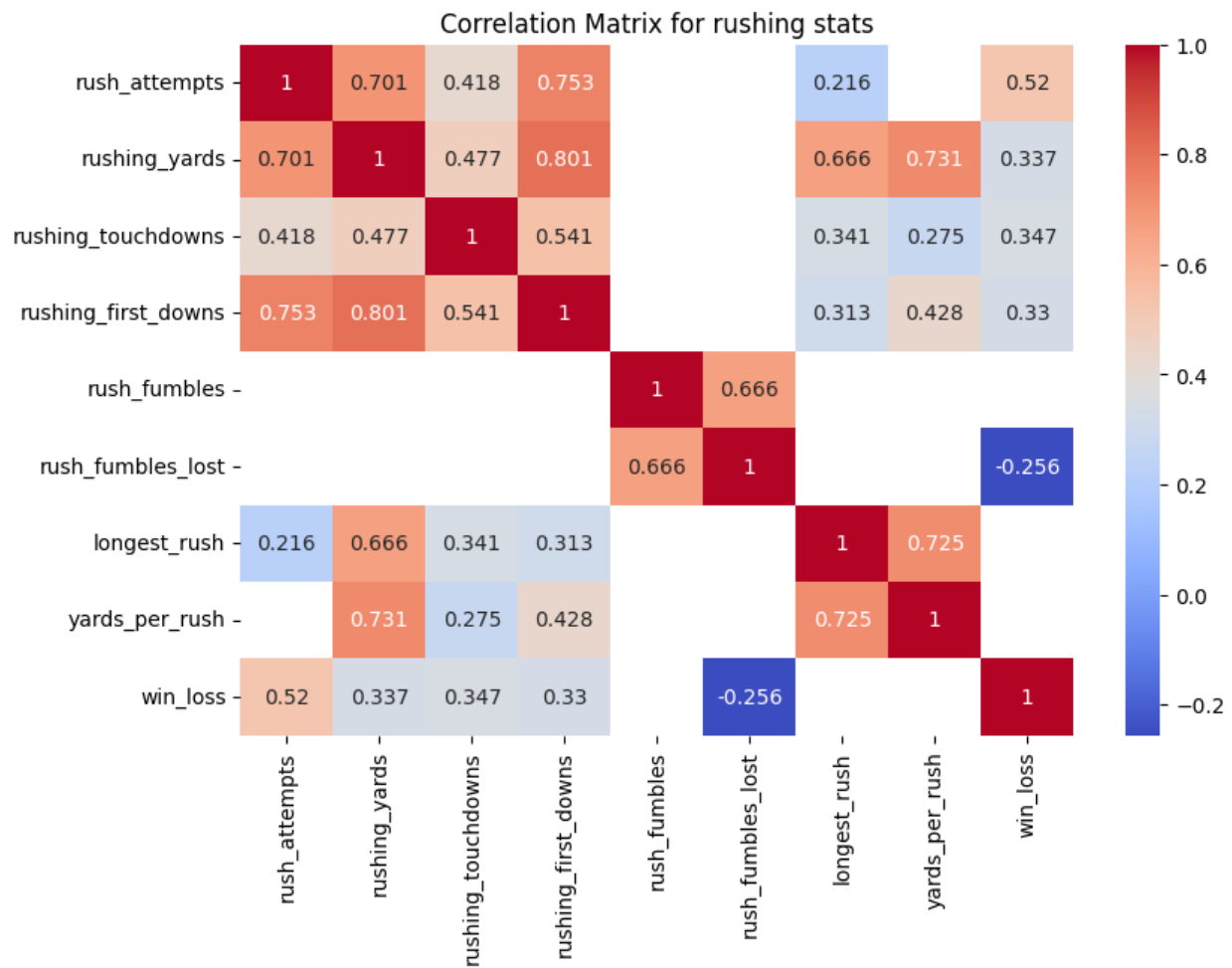
The charts above show extra points made and extra points attempted similarly to the field goals made and attempted the loss class is vastly more prevalent for extra points made and attempted one and below.

Figure 10: Heatmap for Passing Stats



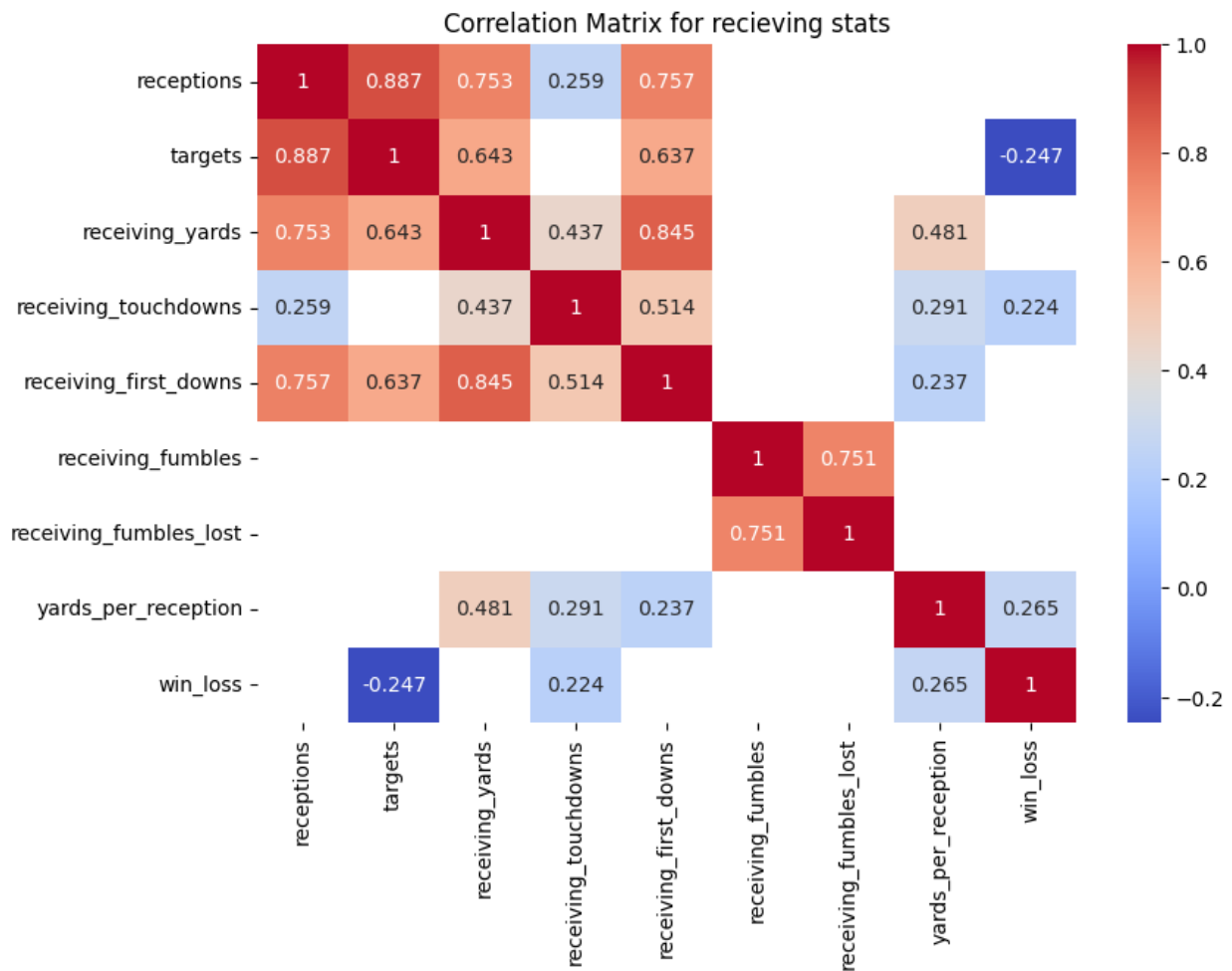
The above chart shows the heatmap for the passing statistics. From this we can see a lot of high correlation values. A lot of these make sense as the more sacks you have it would make sense that you would have more sack yards lost.

Figure 11: Heatmap for Rushing Stats



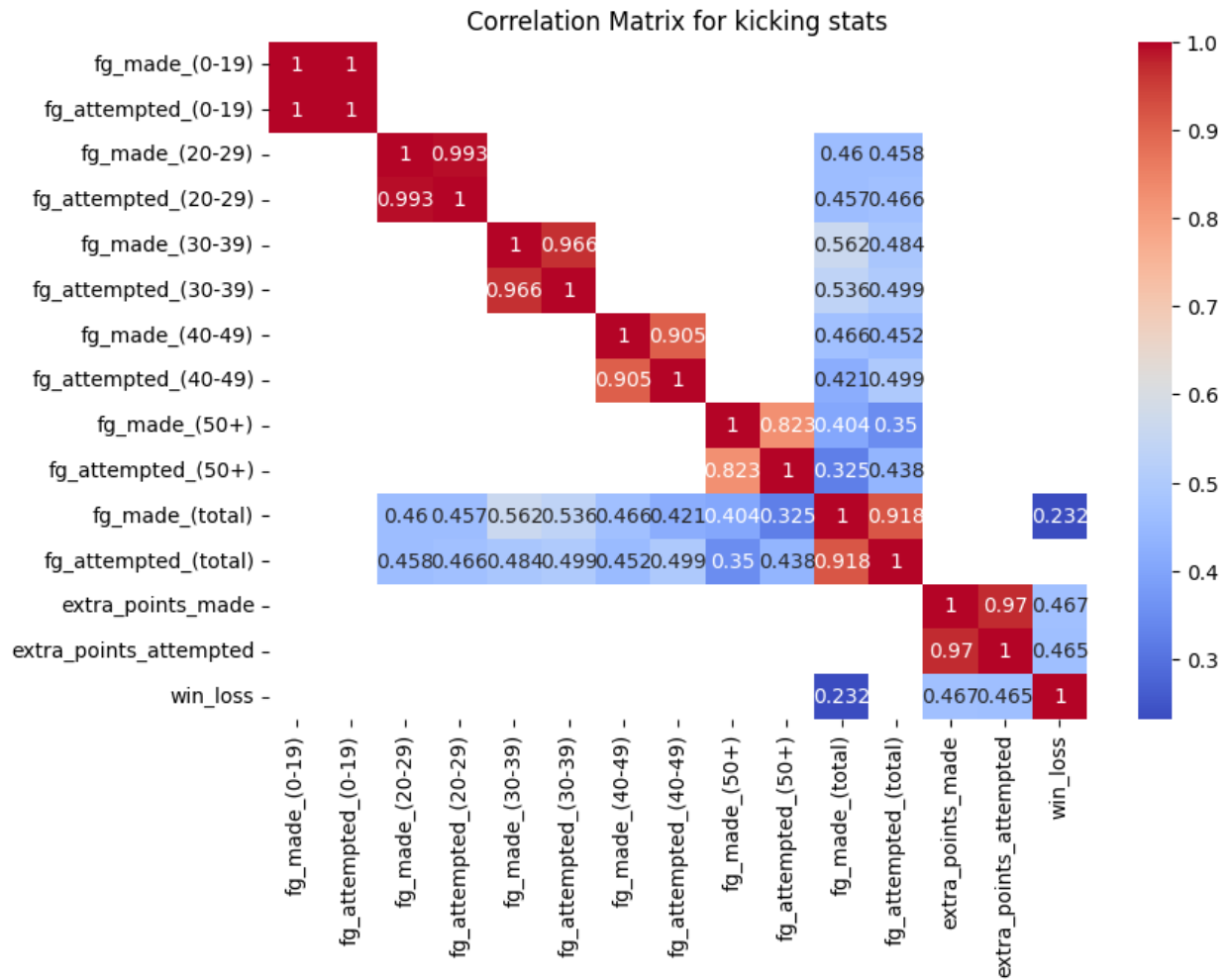
The above chart shows the heatmap for the rushing statistics. From this we can see a lot of high correlation values. Similarly to the passing heatmap, the high correlation makes sense. These are things that I will need to keep in mind during the modeling process so I can avoid multicollinearity.

Figure 12: Heatmap for Receiving Stats



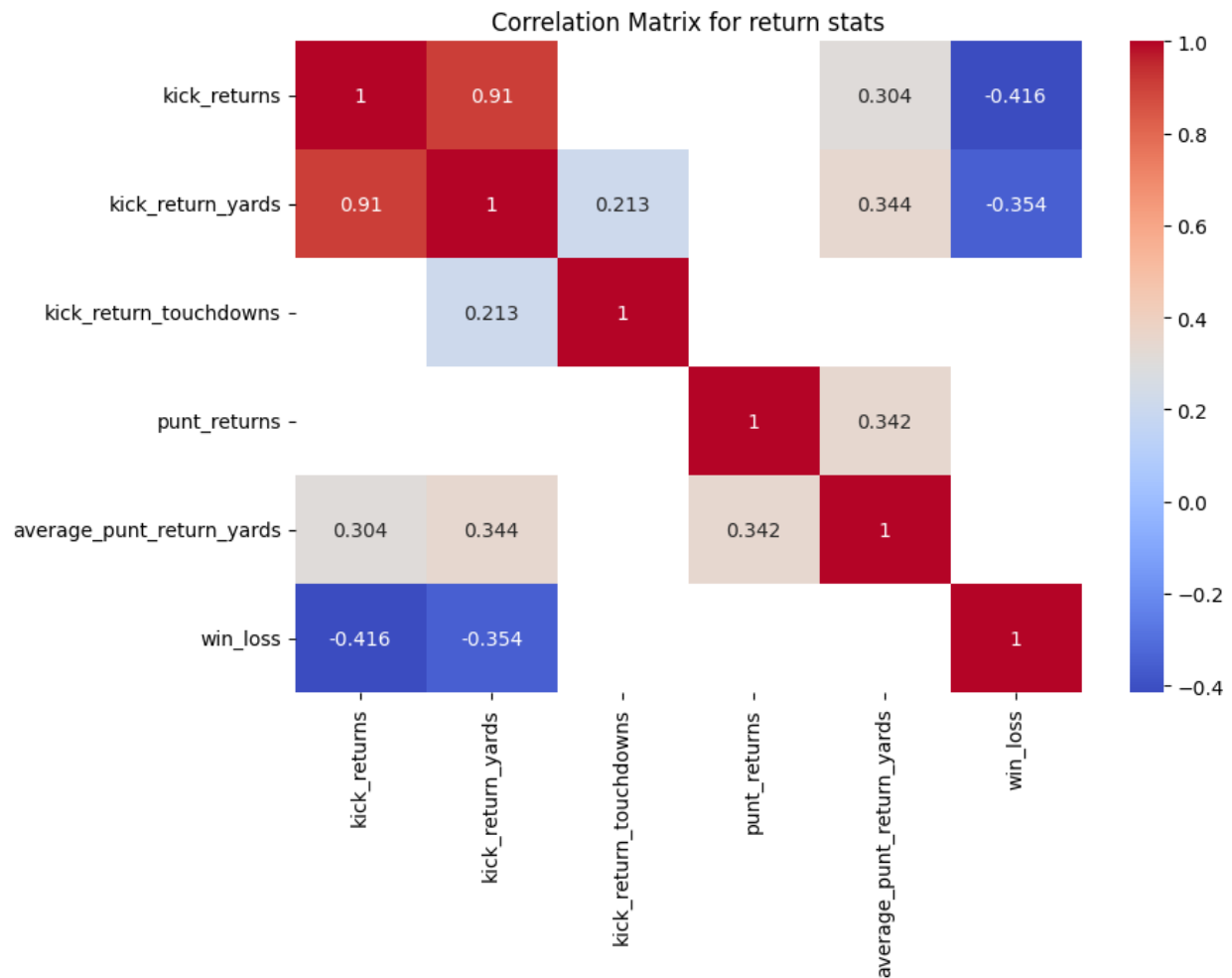
The above chart shows the heatmap for the receiving statistics. From this we can see a lot of high correlation values specifically with the receptions, targets, and first downs columns.

Figure 13: Heatmap for Kicking Stats



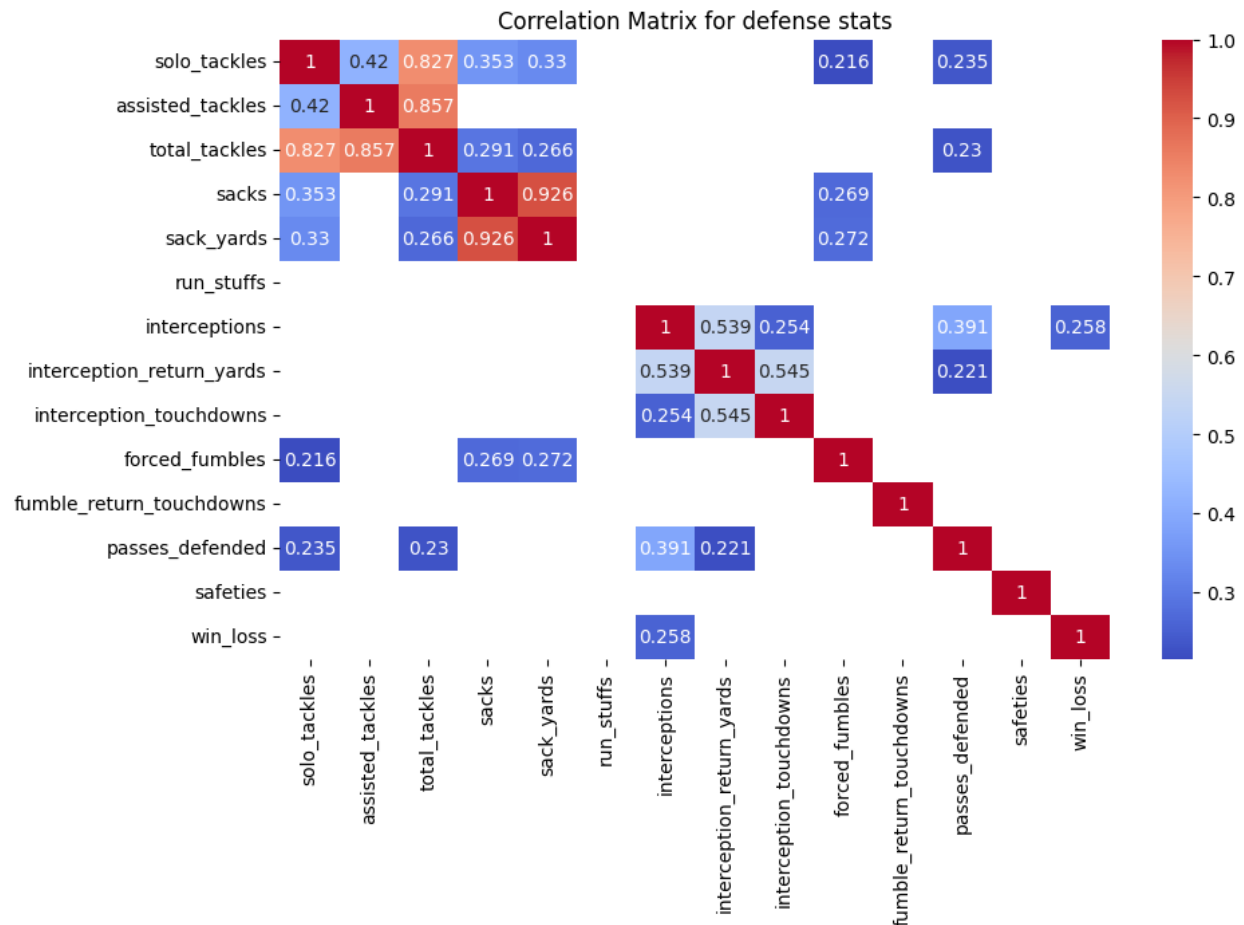
The above chart shows the heatmap for the kicking statistics. From this we can see a lot of high correlation values. Since there is a column for field goals made and attempted, these are going to be very highly correlated since most kickers have a remarkably high make percentage. For modeling it would be smart to engineer a new feature from these or choose one or the other to include in the model.

Figure 14: Heatmap for Return Stats



The above chart shows the heatmap for the return statistics. From this we can see a few high correlation values specifically between kick return yards and kick returns.

Figure 15: Heatmap for Defense Stats



The above chart shows the heatmap for the defense statistics. From this we can see a lot of high correlation values specifically in the sacks and tackles columns.

Modeling

Now that I have familiarized myself with the data and taken notice of high correlation features with the predictors and with the target variable, I feel comfortable starting the modeling process. There were missing values in this dataset so to resolve this I filled them with 0. I decided to do this because for a lot of these missing values the event simply did not occur. For example, if no field goals were attempted from 50+ yards the field would be left empty.

This dataset has a lot of columns so instead of building a model on the entire dataset I decided to build a logistic regression model on each of the statistics groups e.g. One on the

passing columns, one on the rush columns, and so on. After running each model, I would add significant columns to a list to use in my final model. The results of each model can be found below.

Figure 16: Passing Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	373				
Method:	MLE	Df Model:	12				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.2826				
Time:	17:34:50	Log-Likelihood:	-191.94				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	3.221e-26				
	coef	std err	z	P> z	[0.025	0.975]	
const	-0.9969	4.219	-0.236	0.813	-9.266	7.273	
pass_completions	-0.1830	0.246	-0.743	0.457	-0.666	0.300	
pass_attempts	-0.0099	0.131	-0.075	0.940	-0.266	0.247	
passing_yards	-0.0045	0.012	-0.372	0.710	-0.028	0.019	
passing_touchdowns	-0.0305	0.139	-0.219	0.827	-0.303	0.242	
interceptions_thrown	-0.7151	0.169	-4.228	0.000	-1.047	-0.384	
passing_first_downs	0.2911	0.081	3.597	0.000	0.132	0.450	
times_sacked	-0.0848	0.166	-0.512	0.609	-0.409	0.240	
sack_yards_lost	-0.0326	0.024	-1.370	0.171	-0.079	0.014	
pass_fumbles	0.3221	0.242	1.330	0.183	-0.153	0.797	
pass_fumbles_lost	-1.2281	0.406	-3.027	0.002	-2.023	-0.433	
completion_percentage	3.8204	7.913	0.483	0.629	-11.689	19.330	
yards_per_attempt	0.2538	0.361	0.704	0.482	-0.453	0.961	

Figure 17: Rushing Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	377				
Method:	MLE	Df Model:	8				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.3222				
Time:	17:34:57	Log-Likelihood:	-181.36				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	4.077e-33				
	coef	std err	z	P> z	[0.025	0.975]	
const	-5.8326	1.842	-3.167	0.002	-9.442	-2.223	
rush_attempts	0.2767	0.075	3.688	0.000	0.130	0.424	
rushing_yards	-0.0115	0.016	-0.727	0.468	-0.043	0.020	
rushing_touchdowns	0.6327	0.175	3.624	0.000	0.291	0.975	
rushing_first_downs	-0.1782	0.093	-1.916	0.055	-0.360	0.004	
rush_fumbles	-0.0680	0.204	-0.333	0.739	-0.468	0.332	
rush_fumbles_lost	-1.0842	0.322	-3.365	0.001	-1.716	-0.453	
longest_rush	0.0288	0.015	1.867	0.062	-0.001	0.059	
yards_per_rush	0.0454	0.394	0.115	0.908	-0.727	0.817	

Figure 18: Receiving Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	377				
Method:	MLE	Df Model:	8				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1746				
Time:	17:35:41	Log-Likelihood:	-220.82				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	9.240e-17				
	coef	std err	z	P> z	[0.025	0.975]	
const	3.1901	2.065	1.545	0.122	-0.858	7.238	
receptions	-0.1085	0.119	-0.908	0.364	-0.343	0.126	
targets	-0.1580	0.036	-4.383	0.000	-0.229	-0.087	
receiving_yards	0.0118	0.010	1.167	0.243	-0.008	0.032	
receiving_touchdowns	0.1344	0.125	1.078	0.281	-0.110	0.379	
receiving_first_downs	0.2379	0.074	3.218	0.001	0.093	0.383	
receiving_fumbles	-0.2795	0.269	-1.038	0.299	-0.807	0.248	
receiving_fumbles_lost	-0.4092	0.411	-0.997	0.319	-1.214	0.395	
yards_per_reception	-0.1484	0.180	-0.822	0.411	-0.502	0.205	

Figure 19: Kicking Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	374				
Method:	MLE	Df Model:	11				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.2699				
Time:	17:35:47	Log-Likelihood:	-195.33				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	2.036e-25				
	coef	std err	z	P> z	[0.025	0.975]	
const	-3.3497	0.412	-8.126	0.000	-4.158	-2.542	
fg_made_(0-19)	0.0003	nan	nan	nan	nan	nan	
fg_attempted_(0-19)	0.0003	nan	nan	nan	nan	nan	
fg_made_(20-29)	0.6892	0	inf	0.000	0.689	0.689	
fg_attempted_(20-29)	-0.2803	1.08e+07	-2.59e-08	1.000	-2.12e+07	2.12e+07	
fg_made_(30-39)	0.5073	1.873	0.271	0.787	-3.164	4.178	
fg_attempted_(30-39)	-0.2813	1.08e+07	-2.6e-08	1.000	-2.12e+07	2.12e+07	
fg_made_(40-49)	-0.3309	1.529	-0.216	0.829	-3.327	2.665	
fg_attempted_(40-49)	0.1389	1.08e+07	1.28e-08	1.000	-2.12e+07	2.12e+07	
fg_made_(50+)	0.0658	1.583	0.042	0.967	-3.037	3.169	
fg_attempted_(50+)	0.0366	1.08e+07	3.38e-09	1.000	-2.12e+07	2.12e+07	
fg_made_(total)	0.9316	1.626	0.573	0.567	-2.256	4.119	
fg_attempted_(total)	-0.3858	1.08e+07	-3.57e-08	1.000	-2.12e+07	2.12e+07	
extra_points_made	0.2957	0.369	0.802	0.422	-0.427	1.018	
extra_points_attempted	0.7258	0.372	1.951	0.051	-0.003	1.455	

Figure 20: Kickoff Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	382				
Method:	MLE	Df Model:	3				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.2323				
Time:	17:35:56	Log-Likelihood:	-205.40				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	9.165e-27				
	coef	std err	z	P> z	[0.025	0.975]	
const	-3.8232	0.481	-7.951	0.000	-4.766	-2.881	
kickoffs	-0.5393	0.346	-1.560	0.119	-1.217	0.138	
kickoff_yards	0.0214	0.006	3.722	0.000	0.010	0.033	
touchbacks	0.0653	0.129	0.506	0.613	-0.187	0.318	

Figure 21: Punting Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	380				
Method:	MLE	Df Model:	5				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.01309				
Time:	17:36:04	Log-Likelihood:	-264.05				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	0.2202				
	coef	std err	z	P> z	[0.025	0.975]	
const	0.4065	0.242	1.680	0.093	-0.068	0.881	
punts	-0.4747	0.289	-1.643	0.100	-1.041	0.092	
punt_yards	0.0062	0.006	1.085	0.278	-0.005	0.017	
punts_inside_20	0.1542	0.112	1.374	0.169	-0.066	0.374	
punt_touchbacks	0.1120	0.218	0.515	0.607	-0.315	0.539	
blocked_punts	-0.4093	0.884	-0.463	0.643	-2.142	1.324	

Figure 22: Return Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	380				
Method:	MLE	Df Model:	5				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1374				
Time:	17:36:12	Log-Likelihood:	-230.78				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	1.878e-14				
	coef	std err	z	P> z	[0.025	0.975]	
const	2.3508	0.338	6.961	0.000	1.689	3.013	
kick_returns	-0.7337	0.181	-4.062	0.000	-1.088	-0.380	
kick_return_yards	0.0054	0.006	0.831	0.406	-0.007	0.018	
kick_return_touchdowns	1.1431	1.033	1.106	0.269	-0.882	3.168	
punt_returns	-0.1121	0.098	-1.144	0.252	-0.304	0.080	
average_punt_return_yards	0.0020	0.003	0.727	0.467	-0.003	0.007	

Figure 23: Defense Model Results

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	373				
Method:	MLE	Df Model:	12				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1538				
Time:	17:36:19	Log-Likelihood:	-226.41				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	1.517e-12				
	coef	std err	z	P> z	[0.025	0.975]	
const	1.0072	0.410	2.456	0.014	0.204	1.811	
solo_tackles	-0.0280	nan	nan	nan	nan	nan	
assisted_tackles	-0.0090	nan	nan	nan	nan	nan	
total_tackles	-0.0369	nan	nan	nan	nan	nan	
sacks	-0.2126	0.233	-0.913	0.361	-0.669	0.244	
sack_yards	0.0756	0.032	2.398	0.016	0.014	0.137	
interceptions	0.7278	0.253	2.879	0.004	0.232	1.223	
interception_return_yards	0.0277	0.016	1.704	0.088	-0.004	0.060	
interception_touchdowns	-1.9529	0.967	-2.020	0.043	-3.848	-0.058	
forced_fumbles	0.3940	0.208	1.894	0.058	-0.014	0.802	
fumble_return_touchdowns	1.0549	1.202	0.877	0.380	-1.302	3.412	
passes_defended	0.2012	0.073	2.758	0.006	0.058	0.344	
safeties	0.2648	1.366	0.194	0.846	-2.412	2.942	

After all these models ran are their significant column were recorded I was left with 'interceptions_thrown', 'passing_first_downs', 'pass_fumbles_lost', 'rush_attempts', 'rushing_touchdowns', 'rush_fumbles_lost', 'targets', 'receiving_first_downs', 'fg_made_(20-29)', 'kickoff_yards', 'kick_returns', 'sack_yards', 'interceptions', 'interception_touchdowns', and 'passes_defended' to use in my modeling.

Figure 24: Feature Selected Model

Logit Regression Results						
Dep. Variable:	win_loss	No. Observations:	386			
Model:	Logit	Df Residuals:	370			
Method:	MLE	Df Model:	15			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.5364			
Time:	17:43:43	Log-Likelihood:	-124.03			
converged:	True	LL-Null:	-267.55			
Covariance Type:	nonrobust	LLR p-value:	2.750e-52			
	coef	std err	z	P> z	[0.025	0.975]
const	-4.0886	1.434	-2.851	0.004	-6.899	-1.278
interceptions_thrown	-0.5415	0.224	-2.421	0.015	-0.980	-0.103
passing_first_downs	-0.5092	0.545	-0.935	0.350	-1.577	0.558
pass_fumbles_lost	-1.0107	0.620	-1.630	0.103	-2.226	0.205
rush_attempts	0.1375	0.032	4.292	0.000	0.075	0.200
rushing_touchdowns	0.2939	0.193	1.526	0.127	-0.083	0.671
rush_fumbles_lost	-0.2232	0.431	-0.518	0.605	-1.068	0.622
targets	-0.1101	0.036	-3.049	0.002	-0.181	-0.039
receiving_first_downs	0.8015	0.549	1.460	0.144	-0.275	1.878
fg_made_(20-29)	-0.0679	0.278	-0.245	0.807	-0.612	0.476
kickoff_yards	0.0075	0.002	3.545	0.000	0.003	0.012
kick_returns	-0.6198	0.115	-5.376	0.000	-0.846	-0.394
sack_yards	0.0479	0.018	2.713	0.007	0.013	0.082
interceptions	0.9244	0.316	2.924	0.003	0.305	1.544
interception_touchdowns	-1.3758	1.251	-1.099	0.272	-3.828	1.077
passes_defended	0.0952	0.094	1.011	0.312	-0.089	0.280

The figure to the left shows the model results from the significant models in each of the feature groups. Narrowing down the features by group allowed me to digest this model much easier rather than one model with all features included. But even though these were all significant in their individual groups they are not all significant together. From the p-values we can see there a few features with values greater .05 showing they are not making a significant impact on the model.

The next step was to remove the feature with the highest p-value, re-run the model and repeat those steps until all features were significant.

Figure 25: Significant Feature Model Results

Logit Regression Results						
Dep. Variable:	win_loss	No. Observations:	386			
Model:	Logit	Df Residuals:	376			
Method:	MLE	Df Model:	9			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.5232			
Time:	17:52:17	Log-Likelihood:	-127.57			
converged:	True	LL-Null:	-267.55			
Covariance Type:	nonrobust	LLR p-value:	4.616e-55			
	coef	std err	z	P> z	[0.025	0.975]
const	-4.7289	1.298	-3.644	0.000	-7.273	-2.185
interceptions_thrown	-0.5463	0.220	-2.487	0.013	-0.977	-0.116
pass_fumbles_lost	-1.1304	0.426	-2.653	0.008	-1.965	-0.295
rush_attempts	0.1564	0.029	5.325	0.000	0.099	0.214
targets	-0.1040	0.034	-3.062	0.002	-0.171	-0.037
receiving_first_downs	0.2868	0.071	4.025	0.000	0.147	0.426
kickoff_yards	0.0084	0.002	4.322	0.000	0.005	0.012
kick_returns	-0.5976	0.112	-5.348	0.000	-0.817	-0.379
sack_yards	0.0497	0.017	2.890	0.004	0.016	0.083
interceptions	0.9724	0.284	3.419	0.001	0.415	1.530

The figure to the left shows the results of the model after iteratively removing features with p-values greater than .05.

Now that I have my final feature sets I modeled both of the previous two feature sets using logistic regression, random forest, XGBoost, and native bayes keeping track of the accuracy, false positives, and false negatives.

Figure 26: Modeling Results with Feature Selected Columns

```
Logistic Regression Accuracy: 0.8846153846153846
False Positive Rate: 15.384615384615385
False Negative Rate: 7.6923076923076925

Random Forest Accuracy: 0.8589743589743589
False Positive Rate: 17.94871794871795
False Negative Rate: 10.256410256410255

XGBoost Accuracy: 0.8205128205128205
False Positive Rate: 17.94871794871795
False Negative Rate: 17.94871794871795

Naive Bayes Accuracy: 0.8846153846153846
False Positive Rate: 17.94871794871795
False Negative Rate: 5.128205128205128
```

From the results in the figure to the left you can see each model has a solid accuracy score with similar false positive and negative rates. The best performing model was native bayes with an accuracy score of .88, a false positive rate of 17.9% and a false negative rate of 5.1%.

Figure 14: Modeling Results with All Significant Features

```
Logistic Regression Accuracy: 0.8461538461538461
False Positive Rate: 17.94871794871795
False Negative Rate: 12.82051282051282

Random Forest Accuracy: 0.8205128205128205
False Positive Rate: 17.94871794871795
False Negative Rate: 17.94871794871795

XGBoost Accuracy: 0.8076923076923077
False Positive Rate: 17.94871794871795
False Negative Rate: 20.51282051282051

Naive Bayes Accuracy: 0.8589743589743589
False Positive Rate: 17.94871794871795
False Negative Rate: 10.256410256410255
```

From the results in the figure to the left you can see each model has similar accuracy scores with similar false positive and negative rates. The best performing model was native bayes with an accuracy score of .85, a false positive rate of 17.9% and a false negative rate of 10.2%.

Conclusion and Further Study

In conclusion, I was able to utilize exploratory data analysis to familiarize myself with my data and identify certain trends and commonalities between features that could be an indicator of if an NFL team will win or lose. I was also able to model these features and identify significant features using logistic regression. Finally, I was able to build four models

on two different feature sets that resulted in strong accuracy score with decently low false positive and negative rates.

With more time on this project, I would like to collect more data from the coming regular season weeks. I wanted to use current data, so the one downside is having a relatively small dataset. I would have liked to go back and get data from previous seasons, but I could not find a reliable source of data that broke performance down by week. I would also like to evaluate more models and play around with feature engineering. Since there were a lot of correlated features in this dataset, I can think of a few interactions and ratios I could engineer that would be interesting to evaluate the effect on the model by including them.

References

Data Source:

https://sports.yahoo.com/nfl/stats/weekly/?selectedTable=0&week={%22week%22:5,%22seasonPhase%22:%22REGULAR_SEASON%22}